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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

AI-Powered Predictive Fault Detection and Zero-Downtime Maintenance for Electrical Grids

Team ID: *GEN26-A0H-I024*

Team Name: *NEXORA*

Domain: *#2*

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Presentation Outline

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Problem Statement

Electrical-grid faults can develop through abnormal current, voltage, and other electrical signal patterns. Identifying these patterns early is challenging, especially under noisy and changing operating conditions.

Objective: Develop an affordable AI-based monitoring system that detects potential electrical faults and alerts maintenance teams to support early intervention.

Problem Understanding

- **Who is affected?** Electricity utilities and grid operators, substation and maintenance teams, and renewable-energy operators.
- **What causes the problem?** Complex fault patterns, electrical noise, changing loads, and limited fault data for AI training.
- **What is the impact?** Delayed fault identification, unexpected equipment downtime, higher maintenance costs, and safety risks.

Existing Gaps & Challenges

- **Fault detection:** Developing faults can be difficult to distinguish from normal electrical variations.
- **False alarms:** Noise and changing loads can reduce fault-detection reliability.
- **Deployment challenge:** AI models must work within the processing and memory limits of affordable edge devices.

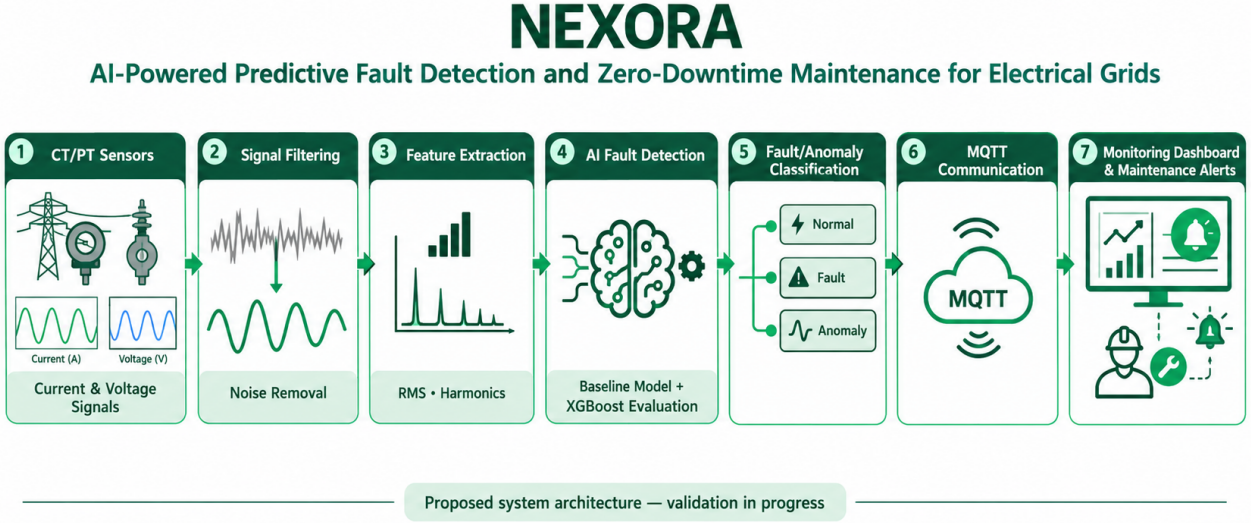
Main challenge: Detect potential faults reliably while keeping the system affordable and practical to deploy.

Our Proposed Solution

- **NEXORA:** An AI-based electrical-grid monitoring system designed to complement existing grid protection systems.
- Collect current and voltage signals, process them to identify important electrical patterns, and use AI to detect anomalies and supported fault types.
- Send alerts to a monitoring dashboard so maintenance teams can prioritise inspections.

Key objective: Provide an additional early-warning layer; NEXORA is not intended to replace existing electrical protection systems.

Technical Approach



CT/PT sensors collect electrical signals. Filtering and feature extraction prepare them for AI analysis. The model identifies potential faults or anomalies, and MQTT sends results to the monitoring dashboard and maintenance alerts.

Tech Stack

Frontend: Web Dashboard

Backend: Python-Based Service

Database: Lightweight Database

AI/ML, APIs, or Hardware: Python, XGBoost, MQTT, CT/PT Sensors,

Innovation & Uniqueness

- **Retrofit approach:** Designed to complement existing grid infrastructure.
- **Edge AI:** Analyses signals locally to reduce dependence on cloud processing.
- **Signal analysis and smart alerts:** Uses electrical features to identify unusual patterns and presents relevant information for maintenance teams.

Key differentiator: A proposed combination of retrofit-oriented monitoring, local AI analysis, and maintenance-focused alerts in one system.

Feasibility & Scalability

- **Feasibility:** Start with a suitable electrical-fault dataset, train a baseline model, compare it with XGBoost, build the dashboard, and test the selected model on suitable edge hardware.
- **Scalability:** Add monitoring nodes to more electrical assets and connect them to a central dashboard after successful testing.
- **Limited data:** Use suitable datasets and document their limitations.
- **False alarms:** Evaluate precision, recall, and false-alarm rate.
- **Hardware and network limits:** Measure processing time and memory; detect missing readings and communication issues.

Cost for Product

Prototype component breakdown (prices checked on 11 October 2026)

Category	Components	Cost (₹)
Sensing	Current sensors, voltage sensors, ADC modules, temperature and humidity sensors	2,313
Edge Processing	ESP32-S3 board, microSD module and 16 GB microSD card	1,499
Power & Wiring	Power supply, fuse, surge protection, resistors, perfboard, terminals and cables	1,249
Communication	MAX485 RS485 (Modbus) module	39
Enclosure & Assembly	IP65 enclosure, mounting accessories, soldering and assembly materials	1,250
Software & Hosting	Cloud VPS for dashboard (3 months)	1,500
Total Prototype Cost		**7,850**
Total with 10% Contingency		**8,635**

TinyML tools and the self-hosted MQTT broker use free tiers. Some component prices are estimates where direct listings were unavailable. Prices may exclude applicable taxes. Optional 4G connectivity and SIM (₹1,799 extra) are not included.

Use Cases

- **Arc-fault pattern detection — Electrical maintenance team:** Analyse electrical signals for patterns associated with suspected arc faults and send an alert for inspection.
- **Solar-grid monitoring — Renewable-energy operator:** Evaluate electrical signals to identify supported fault patterns under changing solar operating conditions.
- **Sensor and communication monitoring — Substation operator:** Identify missing readings or communication interruptions and alert the operator when monitoring reliability is reduced.

Impacts and Benefits

- Earlier visibility into potential electrical faults and better maintenance prioritisation.
- Reduced dependence on manual monitoring alone, supporting reliable electrical infrastructure.
- Evaluate performance using fault-detection precision and recall, false-alarm rate, detection and alert latency, performance under noisy conditions, and edge-device memory usage.

Actual benefits will be established through testing. Reduced downtime is an intended outcome, not yet a proven result.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 7	Target 7.1	Affordable and Clean Energy	Supports reliable access to electricity through improved monitoring of electrical infrastructure.
SDG 9	Target 9.1	Industry, Innovation and Infrastructure	Aims to contribute to reliable and resilient infrastructure through AI-based electrical monitoring.

NEXORA aims to support grid reliability and maintenance through affordable monitoring technology.

Individual Member Contribution

Member	Contribution or responsibility
Kavisri S	AI model and signal processing
Kaushik P	Research and problem analysis
Shruthi Lakshmi V P	Dashboard and software development
Gautham Ashokkumar	Hardware planning and system integration

Update responsibilities if the actual work completed or assigned differs from this draft.

Yukti ID

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